

Assignment No.10

Problem Statement : Design and Develop MongoDB Queries using CRUD operations. (Use CRUD operations, SAVE method, logical operators etc.).

Program :

```
> use Abhi; switched to db Abhi
> db.createCollection('Student');
{ "ok" : 1 }
> db.Student.insert({'Rno':'1','Name':'Piyush','Class':'TE COMP'}); WriteResult({
"nInserted" : 1 })
> db.Student.insert({'Rno':'2','Name':'Abhi','Class':'TE COMP'}); WriteResult({
"nInserted" : 1 })
> db.Student.insert({'Rno':'3','Name':'Ashley','Class':'TE COMP'}); WriteResult({
"nInserted" : 1 })
> db.Student.find();
{ "_id" : ObjectId("5b8fad4ef00832a0a50b5036"), "Rno" : "1", "Name" : "Piyush", "Class" :
"TE COMP" }
{ "_id" : ObjectId("5b8fad62f00832a0a50b5037"), "Rno" : "2", "Name" : "Abhi", "Class" :
"TE COMP" }
{ "_id" : ObjectId("5b8fad70f00832a0a50b5038"), "Rno" : "3", "Name" : "Ashley", "Class" :
"TE COMP" }
> db.Student.find().pretty();
{"_id" : ObjectId("5b8fad4ef00832a0a50b5036"), "Rno" : "1",
"Name" : "Piyush",
"Class" : "TE COMP"}
{"_id" : ObjectId("5b8fad62f00832a0a50b5037"), "Rno" : "2",
"Name" : "Abhi",
"Class" : "TE COMP"}
{"_id" : ObjectId("5b8fad70f00832a0a50b5038"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}
> show dbs; Abhi 0.078GB
admin (empty) local 0.078GB
```

```
> db.Student.update({'Name':'piyush'},{$set:
{'Name':'Hero'}});WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1
})
```

```
> db.Student.find().pretty();
{"_id" : ObjectId("5b8fad4ef00832a0a50b5036"), "Rno" : "1",
"Name" : "Hero",
"Class" : "TE COMP"}
{"_id" : ObjectId("5b8fad62f00832a0a50b5037"), "Rno" : "2",
"Name" : "Abhi",
"Class" : "TE COMP"}
{"_id" : ObjectId("5b8fad70f00832a0a50b5038"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}
```

```
> db.Student.remove({'ADD':'MP'}); WriteResult({ "nRemoved" : 1 })
```

```
> db.Student.find().pretty();
{"_id" : ObjectId("5b8fad62f00832a0a50b5037"), "Rno" : "2",
"Name" : "Abhi",
"Class" : "TE COMP"}
{"_id" : ObjectId("5b8fad70f00832a0a50b5038"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}
```

```
> db.Student.remove({'Name':'Abhi'},1); WriteResult({ "nRemoved" : 1 })
```

```
> db.Student.find().pretty();
{"_id" : ObjectId("5b8fad70f00832a0a50b5038"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}
```

SAVE method and Logical operators use:

```
>db.Student.save({_id:ObjectId("5b8fad4ef00832a0a50b5036"),"RNO
":"1","NAME":"PIYUSH","CLASS":"TE COMP","ADD":"MP"});
```

```
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1})
```

```
> db.Student.find().pretty();
```

```

{"_id" : ObjectId("5b8fad4ef00832a0a50b5036"), "RNO" : "1",
"NAME" : "PIYUSH",
"CLASS" : "TE COMP",
"ADD" : "MP"}
{"_id" : ObjectId("5b8fad70f00832a0a50b5038"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}
> db.Student.find({$and:[{"Name":"Piyush"}, {"Rno":"2"}]});
> db.Student.find({$and:[{"Name":"Piyush"},
{"Rno":"1"}]}).pretty();
{"_id" : ObjectId("5ba1d618f5bbacd4ad81568d"), "Rno" : "1",
"Name" : "Piyush",
"Class" : "TE COMP"}
> db.Student.find({$and:[{"Name":"Piyush"}, {"Rno":"2"}]}).pretty();
> db.Student.find({$or:[ {"Name":"Piyush"}, {"Rno":"2"}]}).pretty();
{"_id" : ObjectId("5ba1d618f5bbacd4ad81568d"), "Rno" : "1",
"Name" : "Piyush", "Class" : "TE COMP"}
> db.Student.find({$or:[ {"Name":"Piyush"}, {"Class":"TE COMP"}]}).pretty();
{"_id" : ObjectId("5ba1d618f5bbacd4ad81568d"), "Rno" : "1",
"Name" : "Piyush",
"Class" : "TE COMP"}
{"_id" : ObjectId("5ba1d63af5bbacd4ad81568f"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}
} db.Student.find( {"Rno": { $not: { $lt: "3" } } }).pretty();
{"_id" : ObjectId("5ba1d63af5bbacd4ad81568f"),
"Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}
{"_id" : ObjectId("5ba1d63af5bbacd4ad81568f"),

```

```

"Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}
> db.Student.find( {"Rno": { $gt:"5"}}).pretty();
{"_id" : ObjectId("5ba1d66df5bbacd4ad815692"), "Rno" : "6",
"Name" : "Pratik",
"Class" : "TE COMP"}
> db.Student.find( {"Rno": { $gte:"5"}}).pretty();
{"_id" : ObjectId("5ba1d65ef5bbacd4ad815691"), "Rno" : "5",
"Name" : "Pratik",
"Class" : "TE COMP"}
{"_id" : ObjectId("5ba1d66df5bbacd4ad815692"), "Rno" : "6",
"Name" : "Pratik",
"Class" : "TE COMP"}
> db.Student.find( {"Rno": { $lt:"5"}}).pretty();
{"_id" : ObjectId("5ba1d618f5bbacd4ad81568d"), "Rno" : "1", "Name" : "Piyush",
"Class" : "TE COMP"}
{"_id" : ObjectId("5ba1d625f5bbacd4ad81568e"), "Rno" : "2",
"Name" : "Abhi",
"Class" : "TE COMP"}
{"_id" : ObjectId("5ba1d63af5bbacd4ad81568f"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}
> db.Student.find( {"Rno": { $lte:"5"}}).pretty();
{"_id" : ObjectId("5ba1d618f5bbacd4ad81568d"), "Rno" : "1",
"Name" : "Piyush",
"Class" : "TE COMP"}
{"_id" : ObjectId("5ba1d625f5bbacd4ad81568e"), "Rno" : "2", "Name" : "Abhi",
"Class" : "TE COMP"}

```

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{"_id" : ObjectId("5ba1d63af5bbacd4ad81568f"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}

{"_id" : ObjectId("5ba1d65ef5bbacd4ad815691"), "Rno" : "5",
"Name" : "Pratik",
"Class" : "TE COMP"}

> db.Student.find( {"Rno": { $lt:"5",$gt:"2"}}).pretty();
{"_id" : ObjectId("5ba1d63af5bbacd4ad81568f"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}

> db.Student.find( {"Rno": { $lte:"5",$gte:"2"}}).pretty();
{"_id" : ObjectId("5ba1d625f5bbacd4ad81568e"), "Rno" : "2",
"Name" : "Abhi",
"Class" : "TE COMP"}

{"_id" : ObjectId("5ba1d63af5bbacd4ad81568f"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}

> db.Student.find( {"Rno": { $lte:"5",$gt:"2"}}).pretty();
{"_id" : ObjectId("5ba1d63af5bbacd4ad81568f"), "Rno" : "3",
"Name" : "Ashley","Class" : "TE COMP"}

{"_id" : ObjectId("5ba1d65ef5bbacd4ad815691"), "Rno" : "5",
"Name" : "Pratik",
"Class" : "TE COMP"}

> db.Student.find( {"Rno": { $lt:"5",$gte:"2"}}).pretty();
{"_id" : ObjectId("5ba1d625f5bbacd4ad81568e"), "Rno" : "2",
"Name" : "Abhi",
"Class" : "TE COMP"}

{"_id" : ObjectId("5ba1d63af5bbacd4ad81568f"), "Rno" : "3",
"Name" : "Ashley",
"Class" : "TE COMP"}

```